

Q1. You are migrating app.devopsbootcamp26.info to an AWS Application Load Balancer.

i . Why can't a standard CNAME record be used at the root (apex) of a DNS zone?

ii. When a DNS resolver queries an AWS ALIAS record, how is the response different from what it would receive from a normal CNAME record?

iii. Do ALIAS records in Route 53 generate DNS query charges? Explain your answer.

Ans:

i. A CNAME record cannot be used at the root (apex) of a DNS Zone because of the DNS protocols rules defined in the RFC standards , the main reason is that zone apex must contain the SOA and NS records. If CNAME is placed at the root , it would violate with the required SOA and NS records.

ii. When DNS resolver queries a normal CNAME record , the DNS server responds as app.devopsbootcamp26.info -> CNAME -> internetfacing-alb.amazonaws.com
So the DNS lookup must resolve the ALB hostname to its A record and then finally returns the IP address. Whereas an ALIAS record is queried directly by the Route 53 as it internally resolve the AWS resource.

iii. Yes ALIAS records in Route 53 generate DNS query charges , but not for the internal AWS resolution to resources

Q2. You have the following requirement:

- EC2 instances inside a VPC must resolve internal.devopsbootcamp26.info to a private IP address.

- Users outside AWS must not be able to resolve this hostname.

1. How would you design this setup using Route 53 hosted zones?

2. If a DNS record exists only in a private hosted zone, but a public hosted zone for the same domain also exists, what happens when:

- an EC2 instance inside the VPC queries the record?
- a user on the public internet queries the same record?

Why can this type of setup cause confusion during incident troubleshooting?

Ans:

1. To design the setup for this i created two hosting zones public which I already have and Private hosted zone and created a subdomain record entry into it to fulfill the requirement.

This is my public hosted zone

The image displays two screenshots of the AWS Route 53 console. The top screenshot shows the 'Public' hosted zone details for 'siriusdevops.online'. It lists the hosted zone name, ID (Z09690572502YU8742UCL), type (Public hosted zone), and name servers. The 'Records' tab shows three records: an NS record for the zone, a SOA record, and a CNAME record for '3tier.siriusdevops.online' pointing to 'tier3-app-alb-690860301.us-east-1.elb.amazonaws.com' with a TTL of 300. The bottom screenshot shows the 'Private' hosted zone details for the same domain. It lists the hosted zone name, ID (Z09690572502YU8742UCL), type (Private hosted zone), and name servers. The 'Records' tab shows three records: an NS record for the zone, a SOA record, and an A record for '3tier.siriusdevops.online' pointing to '44.199.206.231' with a TTL of 300. The right sidebar in the bottom screenshot shows the 'Amazon Q' chat interface.

2. For the given scenario where Public Hosted zone exists , Private Hosted zone exists and record exists only in private zone .

So when for the first scenario where EC2 instance inside VPC queries : the record resolves the domain , because VPC DNS resolver checks Private Hosted Zone first , find the record and return the private IP.

This configuration is known as split horizon DNS. The same domain name returns different results depending on the query source. Internal users receive private IP addresses, while external users cannot resolve the hostname. This can create confusion during incident troubleshooting because DNS appears to work in some environments but fail in others.



- eu-west-1 (disaster recovery)

2. How would you ensure traffic is automatically stopped to a region if the application health check starts failing?

Ans:

1. I would use Latency based routing in route 53 to route users with the lowest latency during normal operation , I will create two DNS records for the same domain one for the ALB in primary region , and the other record pointing to eu-west-1 region. So the Route 53 automatically checks the network latency between the user and AWS regions. Based on this, it routes the user's request to the region that gives the fastest response time.

2. To ensure traffic is automatically stopped to a region if the application health check starts failing , so I would create a health check for each regional endpoint which will be attached to each health check to its respective DNS record. Route 53 if detects a region is unhealthy , will route the traffic automatically.
3. Before Route 53 returns a DNS response, it follows a specific evaluation process, First, it finds all DNS records that match the domain name. Then it checks the health status of each endpoint. Any unhealthy endpoints are removed from consideration. After that, it applies the routing policy (in this case, latency-based routing). Finally, it returns the best healthy endpoint to the user. This process ensures that users are always routed to the lowest-latency region among the healthy application endpoints.

Q4. You are troubleshooting why app.devopsbootcamp26.info does not load in a browser. Run the following command and review the output: dig app.devopsbootcamp26.info.

Based on the output:

1. How do you determine whether the DNS zone exists?
2. How do you determine whether the DNS record exists?
3. What is the practical difference between:
 - NXDOMAIN
 - NOERROR with an empty ANSWER section?
4. What changes would you make to restore DNS resolution, and how would you verify the fix?

Ans:

I will be using my domain name and answer the question based on it , as my app is fully

working , I will just explain how can we figure the above issues.

```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ dig 3tier.siriusdevops.online  
  
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> 3tier.siriusdevops.online  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 58670  
;; flags: qr aa rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 65494  
;; QUESTION SECTION:  
;3tier.siriusdevops.online.      IN      A  
  
;; ANSWER SECTION:  
3tier.siriusdevops.online. 0      IN      A      18.164.237.111  
  
;; Query time: 0 msec  
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)  
;; WHEN: Thu Feb 12 19:40:05 IST 2026  
;; MSG SIZE rcvd: 70  
  
piyush@CK-piyush:~$
```

1. So I am getting status : NOERROR because my app is working and its DNS zone exists , If the status was NXDOMAIN then it would have shown that DNS zone does not exists , so for example I run `dig test.siriusdevops.online` , as this does not exist in route 53 so its status should be NXDOMAIN.

```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ dig test.siriusdevops.online  
  
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> test.siriusdevops.online  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 62086  
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 65494  
;; QUESTION SECTION:  
;test.siriusdevops.online.      IN      A  
  
;; AUTHORITY SECTION:  
siriusdevops.online.      900      IN      SOA      ns-1426.awsdns-50.org. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400  
  
;; Query time: 54 msec  
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)  
;; WHEN: Thu Feb 12 19:43:11 IST 2026  
;; MSG SIZE rcvd: 138  
  
piyush@CK-piyush:~$
```

2. To check if whether DNS record exists , I check the answer section of the output , if the answer section is empty then it means the hosted zone exists but the record is missing. So the answer section confirms whether the record is created or not.
3. The practical difference between NXDOMAIN and NOERROR with empty answer section is :
NXDOMAIN : It means that the domain or hosted zone does not exist and DNS cannot find the domain at all , so the DNS resolution will fail.
NOERROR : It means that the domain or hosted zone does exist but the specific DNS record is missing , so the DNS works , but the required record is not configured.
4. To fix this issue : First I will see that whether the hosted zone exists or not and the NS records are configured correctly . Then I would fix the record if the record exist or will create a new record if it doesn't exist . After verifying the changes I will run the dig command again and will get the status of NOERROR and the answer section populated with the DNS zone and the resolution is completed.

Q5. You updated a DNS record, but users are reporting inconsistent results even after some time has passed.

Compare the output of: dig app.devopsbootcamp26.info

dig @8.8.8.8 app.devopsbootcamp26.info

- 1. What differences do you observe?**
- 2. Where do you find the TTL value in the output, and how does it affect DNS propagation?**
- 3. How would you plan DNS changes in a production environment to avoid this situation in the future?**

Ans:

The output of `dig 3tier.siriusdevops.online` command is :

```
piyush@CK-piyush: ~  
piyush@CK-piyush:~$ dig 3tier.siriusdevops.online  
  
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> 3tier.siriusdevops.online  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 11468  
;; flags: qr aa rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 65494  
;; QUESTION SECTION:  
;3tier.siriusdevops.online.      IN      A  
  
;; ANSWER SECTION:  
3tier.siriusdevops.online. 0      IN      A      18.164.237.111  
  
;; Query time: 0 msec  
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)  
;; WHEN: Thu Feb 12 19:57:48 IST 2026  
;; MSG SIZE rcvd: 70  
  
piyush@CK-piyush:~$
```

The output of `dig @8.8.8.8 3tier.siriusdevops.online` command is :

```
piyush@CK-piyush:~$ dig @8.8.8.8 3tier.siriusdevops.online  
  
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @8.8.8.8 3tier.siriusdevops.online  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47496  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 512  
;; QUESTION SECTION:  
;3tier.siriusdevops.online.      IN      A  
  
;; ANSWER SECTION:  
3tier.siriusdevops.online. 300 IN      CNAME  tier3-app-alb-690860301.us-east-1.elb.amazonaws.com.  
tier3-app-alb-690860301.us-east-1.elb.amazonaws.com. 60 IN A 3.213.245.117  
tier3-app-alb-690860301.us-east-1.elb.amazonaws.com. 60 IN A 54.174.72.147  
  
;; Query time: 68 msec  
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)  
;; WHEN: Thu Feb 12 19:59:31 IST 2026  
;; MSG SIZE rcvd: 151  
  
piyush@CK-piyush:~$
```

1. The first dig command output gave a single A record and IP address , so this output was returned by the local resolver using a locally cached record. The second dig command returned 3 records first being a CNAME record , then two IPs because ALB resolves to multiple IPs to load balance the requests.
2. The TTL value is located in the answer section of the command's output.
It was 0 for local dns resolver and 300 for CNAME record and 60 for ALB IPs. TTL tells DNS resolvers how long they should cache a record. So if TTL is high users may

continue to see old DNS records until the TTL expires. So this is the main reason DNS updates are not reflected immediately anywhere.

3. To avoid inconsistent DNS results in the future, I would first lower TTL before making changes at least 24-48 hours so that caches refresh faster. Then wait for old TTL to expire to ensure all DNS resolvers start using the new lower TTL. Next I would update the DNS record to point to domain to a new a load balancer or IP address, then verify using multiple DNS servers and then increase the TTL again.

Q6. You requested an ACM wildcard certificate for *.devopsbootcamp26.info, but it does not appear when you try to attach it to a CloudFront distribution.

1. What is the most likely reason the certificate is not visible?

2. Why does CloudFront enforce this requirement from an AWS architecture perspective?

3. What steps would you take to resolve the issue and confirm HTTPS is working correctly?

Ans:

1. The most likely reason is that the certificate is created in the wrong AWS region. CloudFront can only use ACM certificate that are created in the us-east-1 region. If the region differs so even if the certificate exists, CloudFront cannot see or use it.
2. CloudFront is a global service. It distributes content through edge locations all over the world. From an AWS architecture perspective:
 - SSL/TLS termination for CloudFront happens at global edge locations.
 - The certificate control plane is managed centrally from us-east-1.
 - Keeping certificates in one region simplifies distribution, replication, and security management.So AWS requires all CloudFront certificates to be stored in us-east-1 to maintain a consistent global architecture.
3. To fix the issue I will follow the following steps :
 - i. I would switch the region first to us-east-1.
 - ii. Then I would request a new ACM certificate for *.devopsbootcamp26.info and then complete the certificate validation and wait until the certificate status becomes 'issued'.
 - iii. Then attach certificate to CloudFront by going to distribution settings and then deploy and wait for the distribution update.
 - iv. Then verify using `https://app.devopsbootcamp.info`, it should work.

Q7. A security scan reports that your application is accessible over plain HTTP.

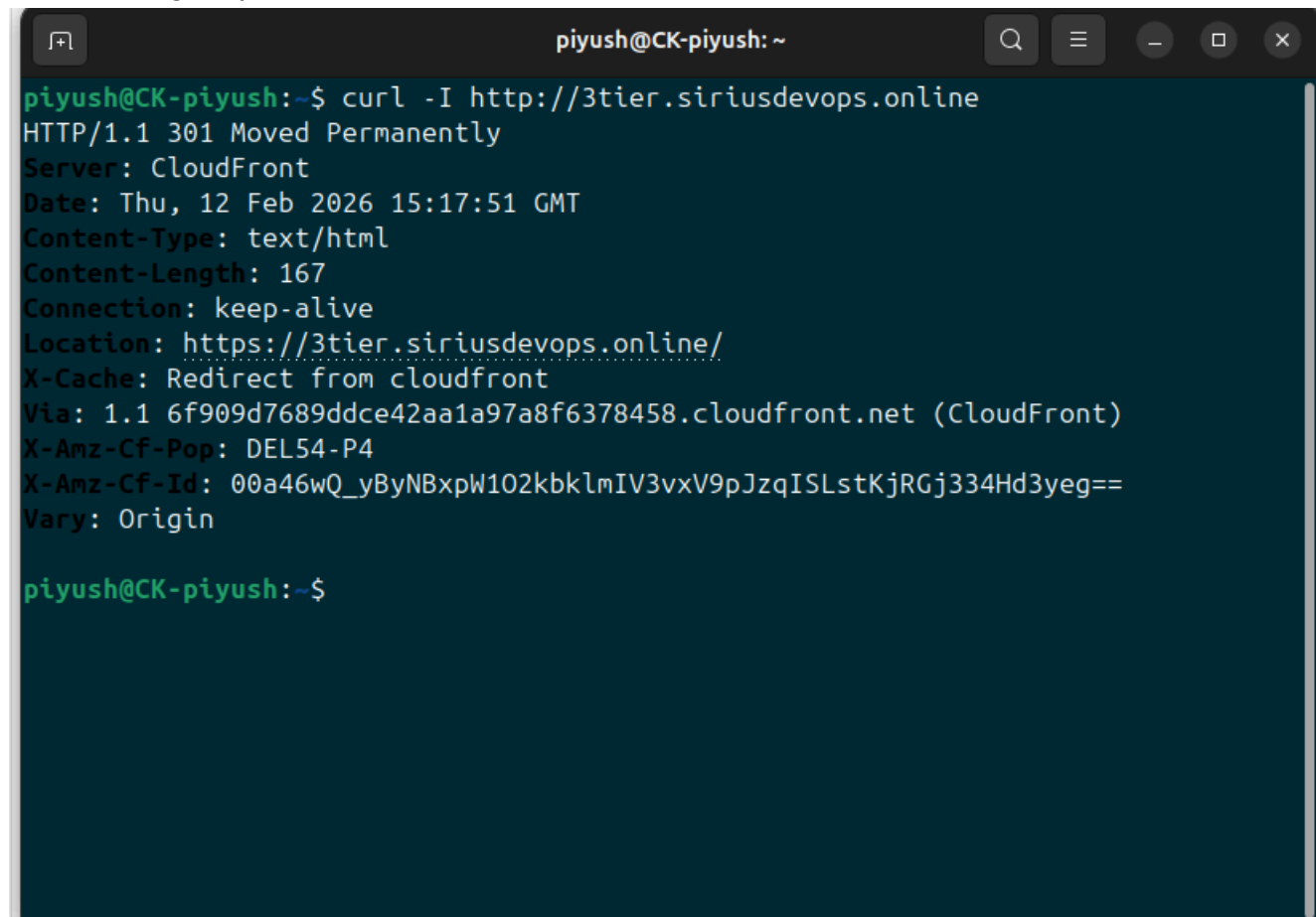
Use the following command and inspect the response: `curl -I http://app.devopsbootcamp26.info`

- 1. What should you look for in the headers?**
- 2. What is most likely misconfigured in the Application Load Balancer?**

How would you fix the issue, and why is a 301 redirect preferred over other status codes?

Ans:

I ran the command for my domain `curl -I http://3tier.siriusdevops.online` and got the following response.

A terminal window titled 'piyush@CK-piyush: ~' with standard window controls. The command 'curl -I http://3tier.siriusdevops.online' is entered. The output shows an HTTP 301 status code with various headers including 'Server: CloudFront', 'Date', 'Content-Type', 'Content-Length', 'Connection', 'Location' (pointing to https://3tier.siriusdevops.online/), 'X-Cache', 'Via', 'X-Amz-Cf-Pop', 'X-Amz-Cf-Id', and 'Vary'.

```
piyush@CK-piyush:~$ curl -I http://3tier.siriusdevops.online
HTTP/1.1 301 Moved Permanently
Server: CloudFront
Date: Thu, 12 Feb 2026 15:17:51 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Location: https://3tier.siriusdevops.online/
X-Cache: Redirect from cloudfront
Via: 1.1 6f909d7689ddce42aa1a97a8f6378458.cloudfront.net (CloudFront)
X-Amz-Cf-Pop: DEL54-P4
X-Amz-Cf-Id: 00a46wQ_yByNBxpW102kbklmIV3vxV9pJzqISLstKjRGj334Hd3yeg==
Vary: Origin

piyush@CK-piyush:~$
```

1. In the response headers , I will check whether my HTTP traffic is being redirected to HTTPS . So it should return the status code as 301 and pointing to the HTTPS url . If the domain was running on HTTP , then the response would be HTTP/1.1 200 OK showing that application is still running on HTTP only and it is not secure.
2. The most likely issue is in the ALB listener configuration.Possible misconfigurations include:

- No HTTP listener on port 80
 - HTTP listener forwarding traffic to target group instead of redirecting
 - Missing redirect rule from HTTP -> HTTPS
- So the ALB is allowing plain HTTP traffic instead of forcing secure HTTPS.

3. To fix the issue , I would add a listener on ALB to HTTPS port 443 and then configure the listener port HTTP :80 to redirect to port 443. After saving , all HTTP requests will automatically redirect to HTTPS.

The 301 (Moved Permanently) status code is preferred over other status code because it tells browser the redirect is permanent , because the browser cache the redirect improving the performance. It enforces secure access consistently.